

SRT LAB

JAILBREAK EDITION

FCA / STELLANTIS

An ECU module workbench you can actually run from a browser tab.

Read VINs. Patch bins. Talk UDS. Crack the secure gateway. No cloud, no telemetry, no subscription.

XTEA · SGW UNLOCK

2018+ Secure Gateway · key from CDA.swf

14 · SEED → KEY ALGOS

CDA6 · GPEC1/2/2A/3 · NGC · JTEC · TIPM · SBEC

AUTEL J2534 · HTTP BRIDGE

MaxiFlash VCI · raw PassThru when serial isn't enough

ON-BENCH · BIN PATCHER

GPEC2A · RFHUB · BCM · 95640 — auto-detect & CRC

01 · LIVE OBD & UDS

Talk to the car.

- **ELM327 / STN**
auto-init (ATZ 3000 ms · STN PP2C/PP2D), one-click reconnect
- **Multi-module scan**
BCM · RFHUB · ECM · TCM · ADCM · IPC · ABS · TIPM · SGW
- **Raw UDS console**
type 22 F1 90, get bytes back, NRC names decoded inline
- **Unlock chain**
preferred algo first, then walks the fallback list on NRC 0x35
- **Response-pending aware**
handles 0x78 polling without dropping the session

WEB SERIAL

02 · VIN SURGERY

Write the VIN. Verify it stuck.

- **Six DIDs covered**
F190 · 7B90 · 7B88 · 6E2025 · 6E2027 · 6EF190
- **24-bit DID encoding**
0x6E_____ space encoded as 3 bytes, no silent truncation
- **Per-module DID list**
BCM gets 6E2025, RFHUB gets 6E2027, EPS gets 6EF190
- **Tail-8 comparator**
accepts either full 17 or trailing-8 on mirror DIDs
- **Write-gating**
refuses 2E if unlock failed, no half-flashed modules

PER-DID READ-BACK

03 · MODULE BIN WORKBENCH

Drag in a dump. Get the truth.

- **Auto-detect**
GPEC2A · RFHUB Gen1/Gen2 · BCM D-FLASH · 95640 EEPROM
- **CRC primitives**
CCITT-FALSE 0x1021 · 95640 0x42 · RFHUB 0xA0 reflected
- **VIN patch with CRC**
patches every slot, recomputes every checksum
- **Virginizer · SKIM toggle · ZZZZ tamper check · transponder & secret keys**
- **IMMO backup sync**
mirrors 0x40C0 → 0x2000 (192 B), 24-byte SKIM records

CLIENT-SIDE PARSE

04 · SECURITY & SEED→KEY

Fourteen algos. One XTEA.

- **XTEA SGW**
128-bit key from CDA.swf, 32 rounds, 4-byte and 8-byte seeds
- **Per-module pref**
ECM/TCM/DAMP/ADCM → GPEC2; body bus → CDA6; SGW → XTEA
- **ADCM Routine 0x0312**
primary unlock with SBEC fallback on rejection
- **RFH ↔ PCM pairing**
vehicle secret cross-check, byte-reversed endian
- **Cross-vehicle key matcher**
sync VINs + keys from a chosen source module

NRC-AWARE CHAIN

05 · HARDWARE & WORKFLOWS

Cable in. Coffee on. Go.

- **Autel MaxiFlash J2534**
local HTTP bridge daemon for SGW-routed writes
- **Program-All**
guided BCM → RFHUB → ECM → ADCM with cross-verify gates
- **On-bench mode**
full UDS over a bench harness, no car required
- **Backups (50) + Sessions (500)**
per-write paper trail with PDF export
- **Jailbreak presets**
SRT · Demon · Hellcat · Redeye feature unlocks

BENCH OR IN-CAR

06 · 18 TABS · ONE TAB IN YOUR BROWSER

Everything, in plain sight.

- **PROGRAM ALL · BCM · RFHUB · ECM · ADCM**
the daily driver flow
- **UDS PROGRAMMER · BACKUPS · SESSIONS · JAILBREAK**
power-user surface
- **DUMPS · BENCH · SEED→KEY · GPEC · GPEC2A**
bench & bin work
- **FCA ANALYZER · LIVE OBD · SWARM · J2534**
diagnostic & hardware
- **Inline styles · React 18 + Vite · pnpm monorepo · Node 24**

NO INSTALL

100% CLIENT-SIDE · NO TELEMETRY · WEB SERIAL + J2534

Open the page. Plug in the cable. The car talks. Your data never leaves the laptop.